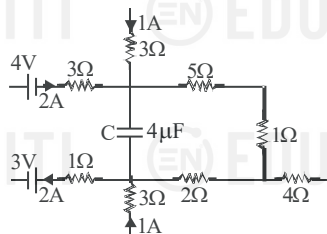
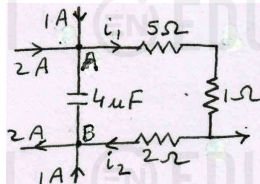


SUBJECTIVE TYPE

1. A part of circuit in steady state along with the currents flowing in the branches, is shown in the figure. Calculate the energy stored in the capacitor, ($C = 4\mu\text{F}$)



1.



In steady state, no current flows through the capacitor. Applying Kirchhoff's junction rule to junctions A and B shown in figure, we get $i_1 = 3\text{A}$ and $i_2 = 1\text{A}$

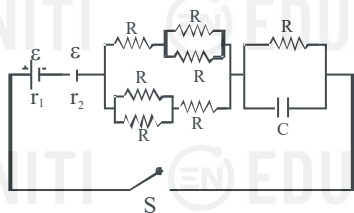
The potential difference across capacitor is

$$V = V_A - V_B = (5 + 1)i_1 + 2i_2 = 6 \times 3 + 2 \times 1 = 20\text{V}.$$

The energy stored in capacitor is

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \times 4 \times 10^{-6} \times 20^2 = 8 \times 10^{-4} \text{ J} = \boxed{0.8\text{mJ}}$$

2. In the circuit shown in the figure, initially switch S is open and capacitor is uncharged. Internal resistances of the cells are r_1 and r_2 and their emf's are equal to ε . The potential difference across the cell of internal resistance r_1 becomes zero long time after closing the switch. Find the value of R in terms of other known physical quantities. All symbols have their usual meaning.



2. In steady state, no current flows through the capacitor. So, capacitor can be removed to find the current in steady state. Now, the resistance of the network of resistors of resistances R is

$$R' = \frac{1}{2} \left(R + \frac{R}{2} \right) + R = \frac{7R}{4}$$

$$\text{The current in the circuit is } i = \frac{\varepsilon + \varepsilon}{r_1 + r_2 + 7R/4} \quad \dots(1)$$

As the potential difference across the cell of internal resistance r_1 is zero, we have $\varepsilon - ir_1 = 0$ (2)

$$\text{From (1) and (2), } \frac{\varepsilon}{i} = \frac{1}{2} \left(r_1 + r_2 + \frac{7R}{4} \right) = r_1$$

$$\therefore R = \frac{4}{7} (r_1 - r_2)$$

3. A parallel plate capacitor is filled with a dielectric whose relative permittivity varies with the applied voltage according to the law $\varepsilon_r = aV$, where $a = 1$ per volt. Another capacitor having same dimensions but containing no dielectric is charged to potential 156V and connected in parallel to the first "non-linear" uncharged capacitor. Determine the final voltage across the capacitors.

3. Let C_2 be second capacitor having potential $V_2 = 156\text{V}$ and C_1 be the first capacitor having potential $V_1 = 0$. The first capacitor has same dimensions as the second capacitor but is filled with a dielectric of relative permittivity

$$\varepsilon_r = aV \Rightarrow C_1 = \varepsilon_r C_2 = aVC_2$$

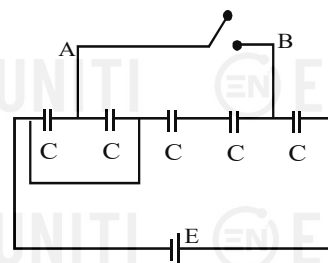
The common potential V across the capacitors when they are connected in parallel is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{0 + C_2 \times 156}{aVC_2 + C_2} = \frac{156}{aV + 1}$$

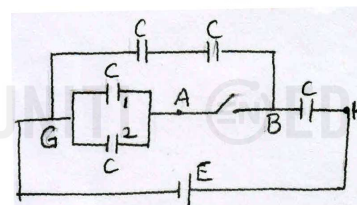
Since $a = 1$, we have $V^2 + V - 156 = 0$

$$\Rightarrow (V + 13)(V - 12) = 0 \quad \therefore \boxed{V = 12\text{V}}$$

4. Find the charge which flows from point A to B, when switch is closed.



4.



The given circuit is redrawn as shown in figure.

When the switch is open, there is no charge on plates 1 and 2.

When the switch is closed, $C_{GB} = 2C + C/2 = 5C/2$

$$\frac{V_{GB}}{V_{BH}} = \frac{C_{BH}}{C_{GB}} = \frac{C}{5C/2} = \frac{2}{5} \quad \text{where } V_{GB} + V_{BH} = E$$

$$\Rightarrow V_{GB} = \frac{2}{2+5} E = \frac{2}{7} E$$

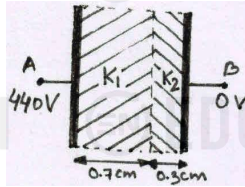
Now, the total charge on plates 1 and 2 is

$$\frac{2}{7} CE + \frac{2}{7} CE = \frac{4}{7} CE$$

Therefore, the charge that flows through the switch is

$$\boxed{\frac{4}{7} CE \text{ from B to A}}$$

5. The plates of a parallel plate capacitor are separated by a distance $d = 1$ cm. Two parallel sided dielectric slabs of thickness 0.7 cm and 0.3 cm fill the space between the plates. If the dielectric constants of the two slabs are 3 and 5 respectively and a potential difference of 440 V is applied across the plates, find
- the electric field intensities in each of the slabs.
 - the ratio of electric energies stored in the first to that in the second dielectric slab.



Given, $K_1 = 3$ and $K_2 = 5$, $d_1 = 0.7$ cm and $d_2 = 0.3$ cm, $V_0 = 440$ V. Let A be the area of plates and σ be the charge density on them.

- (a) The electric field intensities in the two slabs are

$$E_1 = \frac{\sigma}{\epsilon_0 K_1} \text{ and } E_2 = \frac{\sigma}{\epsilon_0 K_2}$$

The potential drops across them are respectively

$$V_1 = \frac{\sigma}{\epsilon_0 K_1} d_1 \text{ and } V_2 = \frac{\sigma}{\epsilon_0 K_2} d_2 \text{ where } V_1 + V_2 = V_0$$

$$\Rightarrow \frac{\sigma}{\epsilon_0 K_1} d + \frac{\sigma}{\epsilon_0 K_2} d = V_0$$

$$\Rightarrow \frac{\sigma}{\epsilon_0} = \frac{K_1 K_2 V_0}{K_2 d_1 + K_1 d_2} = \frac{3 \times 5 \times 440}{5 \times 0.7 \times 10^{-2} + 3 \times 0.3 \times 10^{-2}}$$

$$= 15 \times 10^4 \text{ Vm}^{-1}$$

$$\therefore E_1 = \frac{\sigma / \epsilon_0}{K_1} = \frac{15 \times 10^4}{3} = \boxed{5 \times 10^4 \text{ Vm}^{-1}} \text{ and}$$

$$E_2 = \frac{\sigma / \epsilon_0}{K_2} = \frac{15 \times 10^4}{5} = \boxed{3 \times 10^4 \text{ Vm}^{-1}}$$

- (b) Energy stored in medium
= energy stored per unit volume $\times$ volume

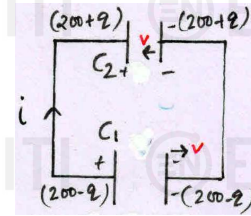
$$= \frac{1}{2} (\epsilon_0 K) E^2 (Ad)$$

$$\therefore \frac{U_1}{U_2} = \frac{K_1 \left(\frac{E_1}{E_2} \right)^2 \frac{d_1}{d_2}}{K_2 \left(\frac{E_2}{E_1} \right)^2 \frac{d_1}{d_2}} = \frac{K_1 \left(\frac{K_2}{K_1} \right)^2 \frac{d_1}{d_2}}{K_2 \left(\frac{K_1}{K_2} \right)^2 \frac{d_1}{d_2}} = \frac{K_2}{K_1} \times \frac{d_1}{d_2}$$

$$= \frac{5}{3} \times \frac{7}{3} = \boxed{\frac{35}{9}}$$

6. A charge $200 \mu\text{C}$ is imparted to each of the two identical parallel plate capacitors connected in parallel. At $t = 0$, the plates of both the capacitors are 0.1 m apart. The plates of first capacitor move towards each other with a constant relative velocity 0.001 m/s and plates of second capacitor move apart with the same constant relative velocity. Find the current in the circuit.

6.



The initial separation between plates is $d = 0.1$ m and the rate of change of separation of plates is $v = 10^{-3}$ m/s.

Let at time t , the charges on capacitors C_1 and C_2 be respectively

$$(200 - q) \mu\text{C} \text{ and } (200 + q) \mu\text{C}.$$

At this instant, the capacitance of the capacitors are

$$C_1 = \frac{A\epsilon_0}{d + vt} \text{ and } C_2 = \frac{A\epsilon_0}{d - vt}$$

Since potential across both the capacitor is same, we have

$$\frac{200 - q}{C_1} = \frac{200 + q}{C_2}$$

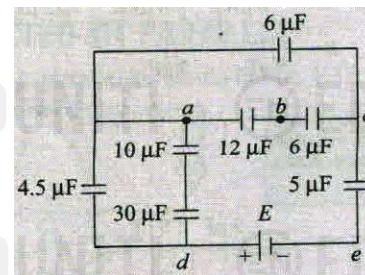
$$\Rightarrow \frac{200 + q}{200 - q} = \frac{C_2}{C_1} = \frac{d + vt}{d - vt}$$

Applying $\frac{N - D}{N + D}$ on both sides, we have $\frac{q}{200} = \frac{vt}{d}$

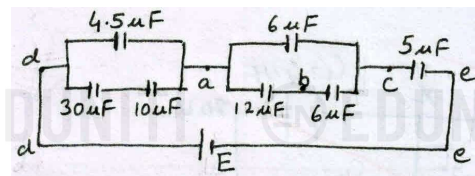
$$\Rightarrow q = \left(\frac{200vt}{d} \right) \mu\text{C}$$

$$\therefore i = \frac{dq}{dt} = \frac{200v}{d} = \frac{200 \times 10^{-3}}{0.1} = \boxed{2 \mu\text{A}}$$

7. In the circuit shown in fig., the potential difference between the points a and b is 4 V. Find the emf of the battery. Assume that before connecting the battery in the circuit, all the capacitors were uncharged.



Solution



We redraw the given circuit as shown in the fig. Two capacitors in series have the same charge and two capacitors in parallel have the same potential difference across them. From fig.,

$$C_{ab} V_{ab} = C_{bc} V_{bc}$$

$$\Rightarrow 12 \times 4 = 6 \times V_{bc}$$

$$\Rightarrow V_{bc} = 8 \text{ V}$$

$$\Rightarrow V_{ac} = V_{ab} + V_{bc} = 4 + 8 = 12 \text{ V}.$$

$$C_{da} = \frac{30 \times 10}{30 + 10} + 4.5 = 12 \mu\text{F} \text{ and } C_{ac} = \frac{12 \times 6}{12 + 6} + 6 = 10 \mu\text{F}$$

$$\frac{1}{C_{de}} = \frac{1}{C_{da}} + \frac{1}{C_{ac}} + \frac{1}{C_{ce}} = \frac{1}{12} + \frac{1}{10} + \frac{1}{5} = \frac{5+6+12}{60}$$

$$\Rightarrow C_{de} = \frac{60}{23} \mu F$$

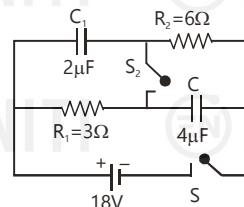
$$Q_{de} = Q_{ac} = C_{ac} V_{ac} = 10 \times 12 = 120 \mu C$$

$$\therefore E = \frac{Q_{de}}{C_{de}} = \frac{120}{60/23} = 46V$$

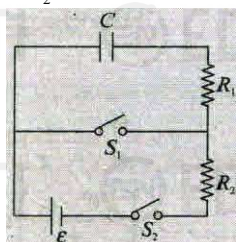
PARAGRAPH TYPE

Paragraph : Q9 to Q11

In the circuit shown, initially the switches are open and capacitors are uncharged. Switches S_1 and S_2 are closed simultaneously at $t = 0$.



8. The capacitor shown in fig. has been charged to a potential difference of V so that it carries a charge CV with both the switches S_1 and S_2 remaining open. Switch S_1 is closed at $t = 0$. At $t = R_1 C$, switch S_1 is opened and S_2 is closed. Find the charge on the capacitor at $t = 2R_1 C + R_2 C$



Solution For $0 < t < R_1 C$, capacitor discharges through R_1 . So,

$$Q = Q_0 e^{-t/R_1} = CV e^{-t/R_1}$$

$$\text{At } t = R_1 C, Q_1 = CV e^{-R_1 C / R_1 C} = CV / e$$

For $R_1 C < t \leq 2R_1 C + R_2 C$, capacitor gets charged through $R_1 + R_2$. Applying Kirchhoff's loop rule to the circuit, we have

$$I(R_1 + R_2) + \frac{Q}{C} - \varepsilon = 0$$

$$\Rightarrow (R_1 + R_2) \frac{dQ}{dt} = \frac{C\varepsilon - Q}{C}$$

If Q_2 is the charge on the capacitor at $t = 2R_1 C + R_2 C$, then

$$\Rightarrow \int_{Q_1}^{Q_2} \frac{dQ}{C\varepsilon - Q} = \frac{1}{(R_1 + R_2)C} \int_{R_1 C}^{(2R_1 + R_2)C} dt$$

$$\Rightarrow \ln \left(\frac{C\varepsilon - Q_2}{C\varepsilon - Q_1} \right) = -\frac{(R_1 + R_2)C}{(R_1 + R_2)C} = -1$$

$$\Rightarrow C\varepsilon - Q_2 = \frac{C\varepsilon - Q_1}{e} = \frac{C\varepsilon - CV/e}{e}$$

$$\therefore Q_2 = C\varepsilon \left(1 - \frac{1}{e} \right) + \frac{CV}{e^2}$$

9. Charge on capacitor C_2 in steady state is

- (a) $12 \mu C$ (b) $24 \mu C$
(c) $48 \mu C$ (d) None of these

10. Now switch S_2 is opened after a long time interval then charge flow through the S_1 is

- (a) $12 \mu C$ (b) $24 \mu C$
(c) $36 \mu C$ (d) $48 \mu C$

11. In above question the amount of heat dissipated in resistors

- (a) $136 \mu J$ (b) $272 \mu J$
(c) $68 \mu J$ (d) $216 \mu J$

9.(c) In steady state, capacitors act as pure conductors. So, the current supplied by battery flows through R_1 , S_2 , R_2 , S_1 and back to battery. The p.d. across C_1 and C_2 are respectively

$$V_1 = \frac{3}{3+6} \times 18 = 6V \text{ and } V_2 = \frac{6}{3+6} \times 18 = 12V$$

The charges stored on C_1 and C_2 are respectively

$$Q_1 = C_1 V_1 = 2 \times 6 = 12 \mu C \text{ and } Q_2 = C_2 V_2 = 4 \times 12 = 48 \mu C$$

10.(d) Before opening switch S_1 , $Q_1 = 12 \mu C$ and $Q_2 = 48 \mu C$. After opening switch S_1 , p.d. across both C_1 and C_2 is $18V$, i.e., $V_1' = V_2' = 18V$. So, the new charges on them are

$$Q_1' = C_1 V_1' = 2 \times 18 = 36 \mu C \text{ and } Q_2' = C_2 V_2' = 4 \times 18 = 72 \mu C$$

The charge that flows through S_1 is

$$\Delta Q = (Q_1' + Q_2') - (Q_1 + Q_2) = (36 + 72) - (12 + 48) = 48 \mu C$$

11.(d) Work done by battery, $W = V \Delta Q = 18 \times 48 = 864 \mu J$

Change in stored energy in capacitors,

$$\begin{aligned} \Delta U &= \left(\frac{1}{2} C_1 V_1'^2 + \frac{1}{2} C_2 V_2'^2 \right) - \left(\frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 \right) \\ &= \left(\frac{1}{2} \times 2 \times 18^2 + \frac{1}{2} \times 4 \times 18^2 \right) - \left(\frac{1}{2} \times 2 \times 6^2 + \frac{1}{2} \times 4 \times 12^2 \right) \\ &= (324 + 648) - (36 + 288) = 648 \mu J \end{aligned}$$

Heat dissipated in resistors,

$$H = W - \Delta U = 864 - 648 = 216 \mu J$$

Paragraph : Q12 to Q13

A sensor to find liquid level in a tank is a cylindrical capacitor of length ℓ with outer and inner conductor radii R_0 and R_i respectively. A liquid fills the tank up to height h , ($h \leq \ell$) from tank's bottom. The dielectric constant of liquid in the lower part is K_l and dielectric constant of vapour of liquid in the upper part is K_v .

12. When liquid fills the tank up to height h , the capacity of capacitor is

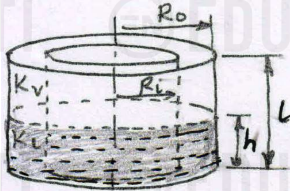
$$\begin{aligned} \text{(a)} & \frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_l - K_v) \frac{h}{\ell} + K_v \right\} \\ \text{(b)} & \frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_v - K_l) \frac{h}{\ell} + K_l \right\} \\ \text{(c)} & \frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_v + K_l) \frac{h}{\ell} + K_v \right\} \\ \text{(d)} & \frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_v + K_l) \frac{h}{\ell} + K_v \right\} \end{aligned}$$

13. If $\ell = 2$ m, $R_0 = 5$ mm, $R_i = 4.5$ mm, $K_l = 1.4$, $K_v = 1.0$, then, capacitance of capacitor when tank is fully filled is

$$\begin{aligned} \text{(a)} & 4.5 \times 10^{-9} \text{ F} & \text{(b)} & 7.5 \times 10^{-9} \text{ F} \\ \text{(c)} & 3.5 \times 10^{-9} \text{ F} & \text{(d)} & 1.5 \times 10^{-9} \text{ F} \end{aligned}$$

Passage I (Solution)

- 12.(a) The two capacitors formed with dielectrics of constant K_l and K_v are in parallel. So equivalent capacitance,



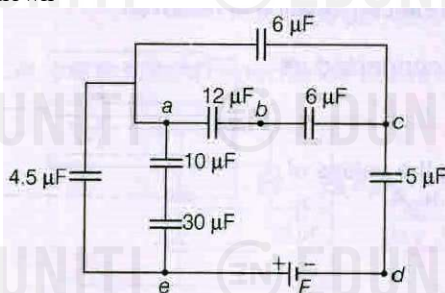
$$\begin{aligned} C_{eq} &= C_l + C_v = \frac{2\pi\epsilon_0 K_l h}{\ln(R_0/R_i)} + \frac{2\pi\epsilon_0 K_v (\ell - h)}{\ln(R_0/R_i)} \\ &= \frac{2\pi\epsilon_0 \ell}{\ln(R_0/R_i)} \left[(K_l - K_v) \frac{h}{\ell} + K_v \right] \end{aligned}$$

- 13.(d) When the tank is full,

$$\begin{aligned} C &= \frac{2\pi\epsilon_0 \ell K_l}{\ln(R_0/R_i)} = \frac{\ell K_l}{2(1/4\pi\epsilon_0) \ln(5/4.5)} \\ &= \frac{2 \times 1.4}{2 \times 9 \times 10^9 \ln(10/9)} \approx 1.5 \times 10^{-9} \text{ F} \end{aligned}$$

Paragraph : Q14 to Q16

A combination of capacitors given is charged by a cell of emf E as shown



If it is given that the potential difference between points a and b is 4 V, then answer the given questions.

14. Potential difference between points a and c is
(a) 4 V (b) 12 V
(c) 10 V (d) 8 V
15. Potential difference between points e and a is
(a) 4 V (b) 12 V
(c) 10 V (d) 8 V
16. EMF E of the charging battery is
(a) 46 V (b) 12 V
(c) 20 V (d) 18 V

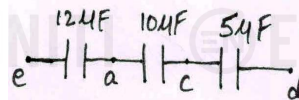
Passage II (Solution)

- 14.(b) In series, voltages across capacitors are in inverse ratio of their capacitances. So,

$$V_{ab}/V_{bc} = 6/12 = 1/2 \Rightarrow V_{bc} = 2V_{ab} = 2 \times 4 = 8V$$

$$\therefore V_{ac} = V_{ab} + V_{bc} = 4 + 8 = 12V.$$

- 15.(c) Circuit is reduced as shown in figure. So,



$$\frac{V_{ea}}{V_{ac}} = \frac{10}{12} = \frac{5}{6}$$

$$\therefore V_{ea} = \frac{5}{6} \times 12 = 10V$$

$$16.(a) \frac{V_{cd}}{V_{ac}} = \frac{10}{5} = 2$$

$$\Rightarrow V_{cd} = 2 \times 12 = 24V$$

$$\therefore E = V_{ea} + V_{ac} + V_{cd} = 10 + 12 + 24 = 46V$$

Paragraph : Q17 to Q19

Two parallel plate capacitors A and B have the same separation ($d = 8.85 \times 10^{-4}$ m) between the plates. The plate area of A and B are 0.04 m² and 0.02 m² respectively. A slab of a medium with dielectric constant $K = 9$ has same dimension such that it can exactly fill the space between the plates of capacitor B.

17. Dielectric slab is put between plates of A and it is charged to a potential of 110 V. Energy stored is

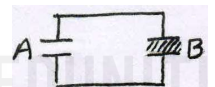
$$\begin{aligned} \text{(a)} & 1.21 \times 10^{-5} \text{ J} & \text{(b)} & 1.21 \times 10^{-3} \text{ J} \\ \text{(c)} & 1.21 \times 10^{-2} \text{ J} & \text{(d)} & 1.21 \times 10^{-6} \text{ J} \end{aligned}$$

18. Battery across A is disconnected and dielectric slab is removed slowly from A. Work done in removing the slab is

$$\begin{aligned} \text{(a)} & 4.84 \times 10^{-6} \text{ J} & \text{(b)} & 6.05 \times 10^{-5} \text{ J} \\ \text{(c)} & 1.21 \times 10^{-5} \text{ J} & \text{(d)} & 4.84 \times 10^{-5} \text{ J} \end{aligned}$$

19. The same dielectric is now put in B and then A and B are connected as shown in fig. Energy stored in the system is

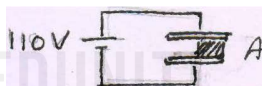
$$\begin{aligned} \text{(a)} & 1.21 \times 10^{-5} \text{ J} \\ \text{(b)} & 6.05 \times 10^{-5} \text{ J} \\ \text{(c)} & 4.84 \times 10^{-5} \text{ J} \\ \text{(d)} & 1.1 \times 10^{-5} \text{ J} \end{aligned}$$



Passage I (Solutions)

- 17.(a) Area of capacitor B and dielectric is $A' = 0.02m^2$ and area of capacitor A is $2A' = 0.04m^2$.

Capacitor A can be treated as combination of two capacitors, one with the dielectric medium and other with air and both in parallel to each other. So,



$$C_A = \frac{A'\epsilon_0 K}{d} + \frac{A'\epsilon_0}{d} = \frac{A'\epsilon_0}{d} (K+1)$$

$$= \frac{0.02 \times 8.85 \times 10^{-12}}{8.85 \times 10^{-4}} \times (9+1) = 2 \times 10^{-9} \text{ F}$$

$$\therefore U_A = \frac{1}{2} C_A V^2 = \frac{1}{2} \times 2 \times 10^{-9} \times (110)^2 = 1.21 \times 10^{-5} \text{ J}$$

- 18.(d) Since the battery is removed, charge Q remains same.

$$Q = C_A V = 2 \times 10^{-9} \times 110 = 2.2 \times 10^{-7} \text{ C}$$

The capacitance of A after removal of slab is

$$C'_A = \frac{2A\epsilon_0}{d} = \frac{2 \times 0.02 \times 8.85 \times 10^{-12}}{8.85 \times 10^{-4}} = 0.4 \times 10^{-9} \text{ F}$$

Energy stored after removal of slab is

$$U'_A = \frac{Q^2}{2C'_A} = \frac{(2.2 \times 10^{-7})^2}{2 \times 0.4 \times 10^{-9}} = 6.05 \times 10^{-5} \text{ J}$$

Work done in removing slab is

$$W = U'_A - U_A = 6.05 \times 10^{-5} - 1.21 \times 10^{-5} = 4.84 \times 10^{-5} \text{ J}$$

- 19.(d) Capacitance of B after insertion of slab is

$$C'_B = \frac{A\epsilon_0 K}{d} = \frac{0.02 \times 8.85 \times 10^{-12} \times 9}{8.85 \times 10^{-4}} = 1.8 \times 10^{-9} \text{ F}$$

Capacitance of system after connecting both capacitors in parallel,

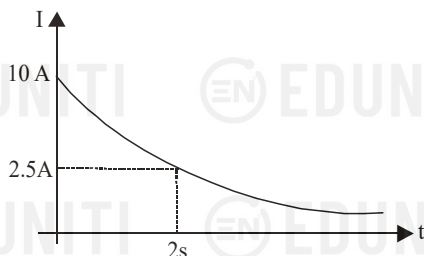
$$C = C'_A + C'_B = 0.4 \times 10^{-9} + 1.8 \times 10^{-9} = 2.2 \times 10^{-9} \text{ F}$$

The total charge on system is Q which only gets redistributed among A and B.

$$\therefore U = \frac{Q^2}{2C} = \frac{(2.2 \times 10^{-7})^2}{2 \times 2.2 \times 10^{-9}} = 1.1 \times 10^{-5} \text{ J}$$

MORE THAN ONE CORRECT

20. The figure shows, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance 10Ω



- (a) The initial potential difference across the capacitor is 100 volt.

(b) The capacitance of the capacitor is $\left(\frac{1}{10 \ln 2}\right) \text{ F}$.

(c) The total heat produced in the circuit will be $\left(\frac{500}{\ln 2}\right) \text{ J}$.

- (d) The thermal power in the resistor will decrease with time whose time constant is $\left(\frac{1}{2 \ln 2}\right) \text{ second}$.

- 20.(a,b,c,d) From graph, $I_0 = 10 \text{ A}$

$$\therefore V_0 = I_0 R = 10 \times 10 = 100 \text{ V}$$

$$\text{For discharging, } I = I_0 e^{-t/RC}$$

From graph, at $t = 2 \text{ s}$, $I = 2.5 \text{ A}$

$$\Rightarrow 2.5 = 10 e^{-2/10C} \Rightarrow 4 = e^{1/5C}$$

$$\Rightarrow \frac{1}{5C} = \ln 4 = 2 \ln 2 \quad \therefore C = \left(\frac{1}{10 \ln 2}\right) \text{ F}$$

Total heat produced = Initial energy stored in capacitor

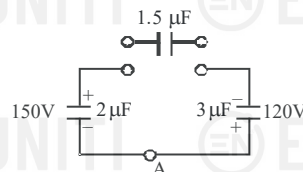
$$= \frac{1}{2} C V_0^2 = \frac{1}{2} \times \frac{1}{10 \ln 2} \times (100)^2 = \left(\frac{500}{\ln 2}\right) \text{ J}$$

$$\text{Power in resistor, } P = I^2 R = (I_0 e^{-t/RC})^2 \times R = P_0 e^{-2t/RC}$$

Therefore, time constant for thermal power

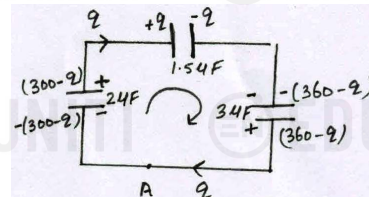
$$= \frac{RC}{2} = 10 \times \frac{1}{10 \ln 2} \times \frac{1}{2} = \frac{1}{2 \ln 2}$$

21. Two capacitors of capacitances $2 \mu\text{F}$ and $3 \mu\text{F}$ are charged to 150 volt and 120 volt respectively. The plates of capacitor are connected as shown in the figure. A discharged capacitor of capacity $1.5 \mu\text{F}$ falls to the free ends of the wire. Then



- (a) charge on the $1.5 \mu\text{F}$ capacitor is $180 \mu\text{C}$
 (b) charge on the $2 \mu\text{F}$ capacitor is $120 \mu\text{C}$
 (c) charge flows through A from right to left
 (d) charge flows through A from left to right.

- 21.(a,b,c)



Initial charges on capacitors are,

$$q_1 = 150 \times 2 = 300 \mu\text{C} \quad \text{and} \quad q_2 = 120 \times 3 = 360 \mu\text{C}$$

Let charge q flows through A from right to left. Then final charge distribution on capacitors is shown in the figure. Applying Kirchoff's voltage law to the loop, in clockwise direction starting from A, we have

$$-\frac{300-q}{2} + \frac{q}{1.5} - \frac{360-q}{3} = 0$$

$$\Rightarrow -3(300-q) + 4q - 2(360-q) = 0$$

$$\Rightarrow 9q = 1620 \quad \therefore q = 180\mu C$$

So charge on $1.5\mu F$ capacitor is $q = 180\mu C$ and on $2\mu F$ capacitor is $300 - 180 = 120\mu C$.

As q is positive, charge flows from right to left through A .

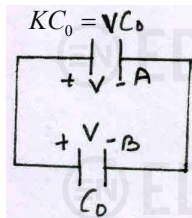
22. A parallel plate capacitor A is filled with a dielectric whose dielectric constant varies with applied voltage as $K = V$. An identical capacitor B of capacitance C_0 with air as dielectric is connected to voltage source $V_0 = 30V$ and then connected to the first capacitor after disconnecting the voltage source.

The charge and voltage on capacitor

(A) A are $25C_0$ and $25V$ (B) A are $25C_0$ and $5V$

(C) B are $5C_0$ and $5V$ (D) B are $5C_0$ and $25V$

22.(b,c)



As shown in figure, let V be the common potential across capacitors. Thus capacitance of capacitor A will be VC_0 . As the total charge before and after connection remains same, we have

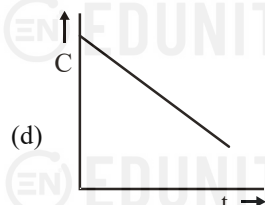
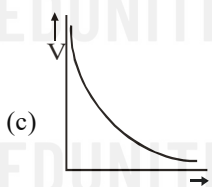
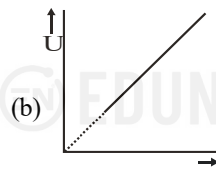
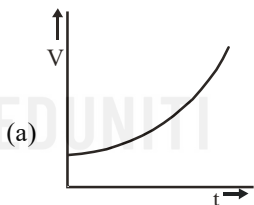
$$C_0 \times 30 = (VC_0 + C_0)V \quad \Rightarrow V^2 + V - 30 = 0$$

$$\Rightarrow (V+6)(V-5) = 0 \quad \therefore V = 5V.$$

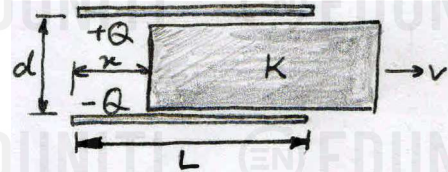
Thus charge on A , $Q_A = (VC_0)V = 25C_0$

and, charge on B , $Q_B = C_0V = 5C_0$

23. A parallel plate capacitor with square plates has a dielectric slab in it. The slab just fills the space inside the capacitor. The capacitor is charged by a battery and then battery is disconnected. Now at $t = 0$, the slab is pulled out slowly with constant speed in the direction parallel to one of the edges of the plates.. If at time t , capacitance of the capacitor is C , potential difference across is V and energy stored in it is U , then which of the following graphs are correct?



23.(a,b,c,d)



Let the dimensions of plates be $L \times L$, distance between them be d and the speed of dielectric be V .

At time t , $x = vt$. So, capacitance at time t is

$$C = \frac{xL\epsilon_0}{d} + \frac{K(L-x)L\epsilon_0}{d} = \frac{L\epsilon_0}{d} [KL - x(K-1)]$$

$$= \frac{L\epsilon_0}{d} [KL - vt(K-1)] \quad \therefore \text{Option (d) is correct.}$$

As charge Q remains constant,

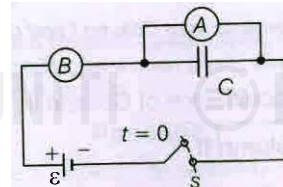
$$V = \frac{Q}{C} = \frac{Qd}{L\epsilon_0 [KL - vt(K-1)]}$$

$\therefore$ Options (a) and (c) are correct.

$$U = \frac{1}{2} QV \quad \Rightarrow U \propto V \text{ as } Q \text{ is constant}$$

$\therefore$ Option (b) is correct.

24. A capacitor of capacitance C is connected to two voltmeters A and B as shown in figure. The resistance of B is R and A is an ideal voltmeter of infinite resistance



The capacitor is initially uncharged and at $t = 0$, the switch is closed.

- (a) At $t = 0$, reading of A will be zero and that of B will be ϵ .
 (b) During time interval $0 \leq t < \infty$, readings of B and A will change
 (c) Readings of A and B will be equal at $t = RC \log_e 2$
 (d) When $t \rightarrow \infty$, reading of A will be ϵ and that of B will be zero.

- 24.(a,b,c,d) Initially at $t = 0$, capacitor is uncharged and can be treated to have zero resistance. So, p.d. across it is zero. Hence, reading of A is zero and that of B is ϵ .

When $t \rightarrow \infty$, capacitor is completely charged and can be treated to have infinite resistance. So, entire p.d. equal to ϵ is across the capacitor and zero across B . Hence, reading of A is ϵ and that of B is zero. At time t , p.d. across capacitor is $V = \epsilon(1 - e^{-t/RC})$

$$\text{At } t = RC \ln 2, V = \epsilon(1 - e^{-\ln 2}) = \epsilon(1 - 1/2) = \epsilon/2.$$

So, reading of A is $\epsilon/2$ and that of B is $\epsilon - \epsilon/2 = \epsilon/2$.

Now, the readings of both A and B are equal.